ACE Board Rev 1.0: Build Notes

The work an ACE Board Rev 1.0 needs beyond its bill of materials, for a new build or for upgrading a built board. Rev 1.1 makes these fixes on the PCB. On Rev 1.0 you make them by hand.

	Change	Who needs it
1	Fit 10 kΩ at R1–R5 and R27, not 1 kΩ	Every board
2	Fit an R6551AP at U4	Every board
3	Keep J1 on its 1 MHz pins	Every board
4	Add a 10 µF supply capacitor on the underside	Every board
5	Program the AB Controller firmware, and add a 10 kΩ pull-up on RESB	Firmware: every board. Pull-up: recommended
6	Stack a socket on U3 and fit a diode for /INT	Any pico9918 video card
7	Fit the ACE RAM Patch Rev 1.1 in place of U12 and U13	Any board with the banked RAM (U15, U21, U22)

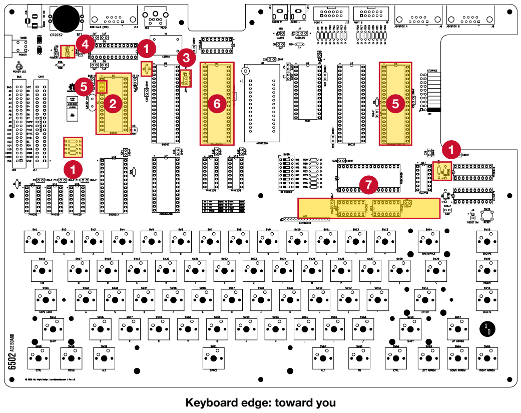


Figure 1. Top of the ACE. Where each change is. Change 5 is in two places: the ATmega, U6, and its pull-up under U4.

Power: use a center-positive 5 V supply only. Rev 1.0 has no reverse-polarity protection, so a reversed supply reaches every chip. Rev 1.1 adds a diode for this. These notes have no hand fix for it.

These notes use Rev 1.0's part numbers

Rev 1.0's designators are the ones printed on your board. Rev 1.1 renumbers many parts, and the bill of materials in the 6502-ACE README uses Rev 1.1's numbers, so build from `Production/ACE Board/Rev 1.0/bom.csv`. Rev 1.1's R5, C20, D1 and U13 are all different parts from the Rev 1.0 parts with those numbers.

Changes to the parts you buy

Part	Qty	For
10 kΩ resistor, 1/8 W	7	Change 1 (R1–R5, R27) and change 5 (RESB). Rev 1.0's BOM lists 34 × 1 kΩ: buy 28 instead
R6551AP	1	Change 2. Already at U4 in the BOM, where the value reads 65C51
10 µF electrolytic, 16 V	1	Change 4
Machined-pin DIP-40 socket, BAT85 diode	1 each	Change 6
ACE RAM Patch Rev 1.1 board, 74HC02	1 each	Change 7
74HC00	4, not 5	Change 7 uses one on the patch, in place of U12 and U13 on the ACE
DIP-14 sockets	2 fewer	Change 7 leaves U12 and U13 without sockets
Pin header, 2.54 mm, four strips of 7 pins	1 set	Change 7

Building a new board? Fit the patch and the underside parts before the keyswitches, sockets and tall parts, while the board still lies flat on its face.

1 Pull-ups: R1–R5 and R27 are 10 kΩ

Rev 1.0's BOM lists 1 kΩ at these six places. Rev 1.1 makes five of them 10 kΩ and removes R1. R2, R5, R3 and R4 pull up shared lines: BE, RDY, NMIB and IRQB. A 1 kΩ pull-up draws 5 mA, more than the 3.2 mA the 65C22's IRQ output is rated to sink. R27 pulls up the banked RAM's chip enable (U15 pin 22). R1, beside U1's top left corner, holds SOB (U1 pin 38) high. Nothing else holds that pin, so don't leave R1 out. Every other 1 kΩ on the board stays 1 kΩ, including R6, which is the ATmega's reset pull-up.

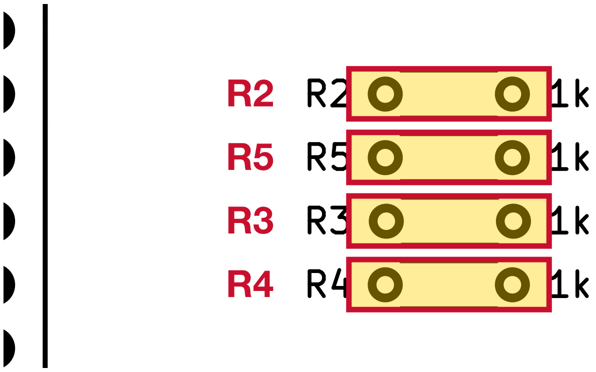


Figure 2. R2–R5, beside the CART connector. Note the order: R2, R5, R3, R4.

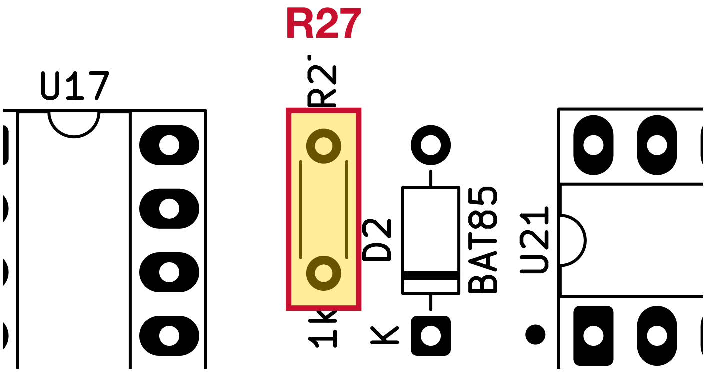


Figure 3. R27, between U17 and U21.

☐ R1, R2, R3, R4, R5: 10 kΩ.

☐ R27: 10 kΩ.

2 U4: fit an R6551AP

The silkscreen and Rev 1.0's BOM say 65C51. The R6551AP is the part the AC6502 machines are tested with. Don't substitute a W65C51N: its transmit-empty status bit does not work.

☐ U4 is an R6551AP.

3 J1: keep the shunt on pins 1 and 2

J1, PHI2 SELECT, offers 1 MHz or 2 MHz. Use 1 MHz. The SID's clock is always 1 MHz, so at 2 MHz the SID answers only every other CPU cycle, and the BIOS's card probes miss cards. Rev 1.1 removes the jumper and runs at 1 MHz only.

☐ Shunt on J1 pins 1 and 2.

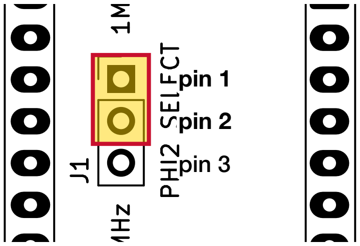


Figure 4. J1: the shunt covers the highlighted pair.

④ Add the 10 µF supply capacitor

Rev 1.1 adds a 10 µF electrolytic across VCC and GND beside the power input, as bulk capacitance for the whole board. Rev 1.0 has no holes for it, so it goes on the underside, across the pads of J19, the VCC header. J19's pads are close to the power switch and 2.54 mm apart, and J19 can still be fitted. Rev 1.0's own C20 is a 100 nF capacitor elsewhere on the board: leave it alone.

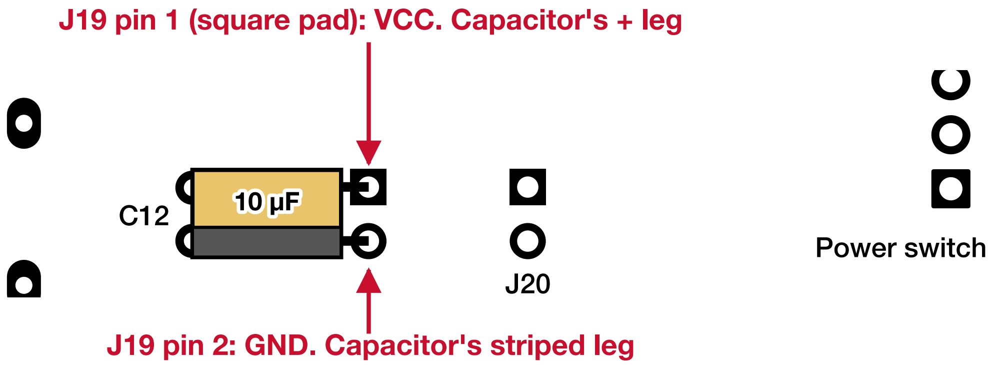

J19 pin 1 (square pad): VCC. Capacitor's + leg
C12
10 µF
J20
Power switch
J19 pin 2: GND. Capacitor's striped leg

Figure 5. The capacitor across J19's pads, lying flat. Underside of the ACE, keyboard edge toward you.

- ☐ Bend the legs to 2.54 mm. Positive leg to J19 pin 1 (VCC), striped leg to J19 pin 2 (GND).
- ☐ Lay the body flat on the board, so it stays clear of the plate. Its insulated sleeve can rest on C12's joints. Solder the legs to the pads, or to J19's tails if J19 is fitted, and clip them.

⑤ Reset: the AB Controller firmware and the RESB pull-up

Rev 1.0 has no pull-up on RESB, and older AB Controller firmware drives the line high as well as low, so nothing else can reset the machine. The firmware in the 6502-ACE repository only ever pulls RESB low. That lets the DS1511Y and any card on CART or BUS reset the machine too.

Program the ATmega

Program U6 out of its socket in a TL866-class programmer, using `minipro` and PlatformIO. In `Firmware/AB Controller/`, for an ATmega1284:

```
pio run -e atmega1284 -t upload
```

Use `-e atmega1284p` for a 1284P. The command also writes the fuses. The firmware's README has the details.

- ☐ U6 programmed, and back in its socket.

Fit the pull-up

The ATmega's internal pull-up raises RESB on its own, but a 10 kΩ is stiffer on a line that runs out to CART and BUS. Rev 1.1 has one. On Rev 1.0, U4's pins 2 and 4 are VCC and RESB, 5.08 mm apart, the same pitch as a resistor footprint.

- ☐ 10 kΩ across U4 pin 2 and U4 pin 4, on the underside. Pin 3 is between them: keep the legs off its joint.

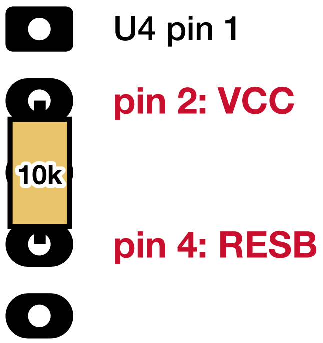


Figure 6. The RESB pull-up on the underside, keyboard edge toward you. U4's pin 1 is nearest the connectors.

6 The /INT diode, on a socket stack at U3

U3 pin 16 is the video chip's /INT, and Rev 1.0 wires it straight to IRQB. A TMS9918A's /INT is open-drain, but every pico9918 drives it high as well as low, and holds IRQB at about 3.8 V while the VIA tries to pull it low. Keyboard and serial input are then dead. This affects the PRO and the v1.3 alike: a v1.3 can appear to work, but only by luck. Rev 1.1 has a diode on the PCB. Rev 1.0's PCB has no place for one, so it goes in a socket stack: a second socket between U3's socket and the card, with pin 16 routed through the diode.

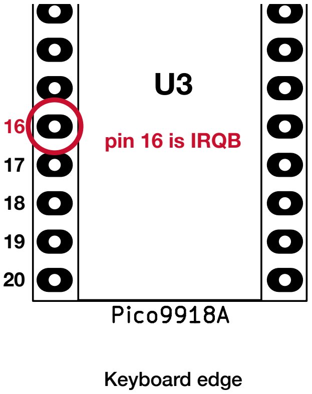
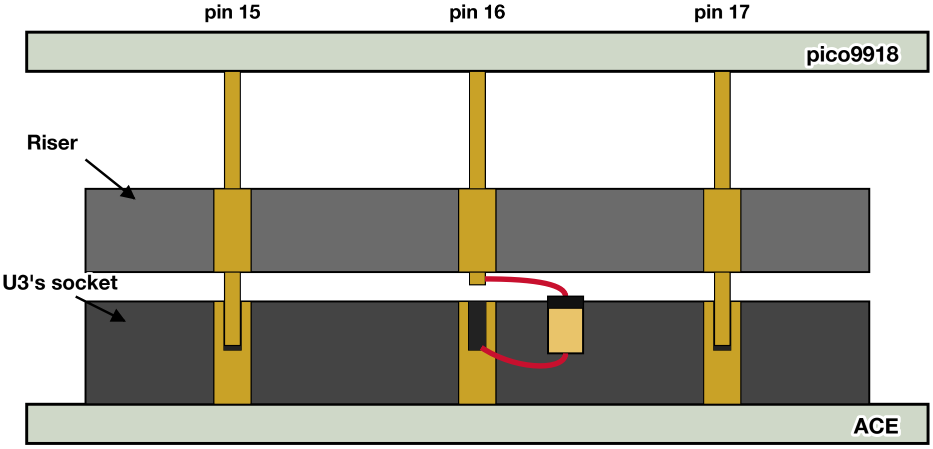


Figure 7. Find pin 16: the fifth pin up from the keyboard end of U3's left row.



Band (cathode) to the riser's pin 16. Other leg (anode) into U3's socket at pin 16, which is IRQB.

Figure 8. The stack, seen from inside the left row. Not to scale. The riser's pin 16 stops short of the socket below, so the only path from the card's pin 16 to IRQB is through the diode.

- ☐ Take a machined-pin DIP-40 socket: this is the riser. The pico9918's pins are solid and can't be bent or cut, so the change is made here.
- ☐ Count to pin 16 on the riser, with its notch the same way as U3's. Cut off pin 16's tail below the plastic, so it can't reach U3's socket.
- ☐ Solder the BAT85's band end (cathode) to what is left of the riser's pin 16.
- ☐ Put the other leg (anode) into U3's socket at pin 16, and press the riser into U3's socket. Keep the diode under the card or on the board beside U3's left row, with its legs clear of the other pins.
- ☐ Check with the meter's diode test, with the card out: red on IRQB (R4's left leg, Figure 2), black in the riser's pin 16. About 0.3 V this way, and open the other way.
- ☐ Fit the card into the riser, with its pin 1 at the connector end. It now sits one socket higher: check that it clears any top panel.

7 The ACE RAM Patch Rev 1.1

Rev 1.0's latches for the banked RAM load at the wrong times, so the RAM doesn't hold a bank. The patch is a small board that replaces U12 and U13. It mounts under the ACE on pin headers, with the same logic as Rev 1.1.

Status: the patch's logic passes `Tests/check-schematics.mjs` in every bus state. Its fit and these steps come from the PCB files, and have not yet been confirmed on a real board. The 6502-ACE README's "ACE RAM Patch" section has the current status.

Before you start

☐ U12 and U13 are empty: no chips and no sockets. On a built board, take them off. Take off an older Rev 1.0 patch or any other rework on this part of the logic.

☐ All 28 holes of U12 and U13 are open and clear of solder.

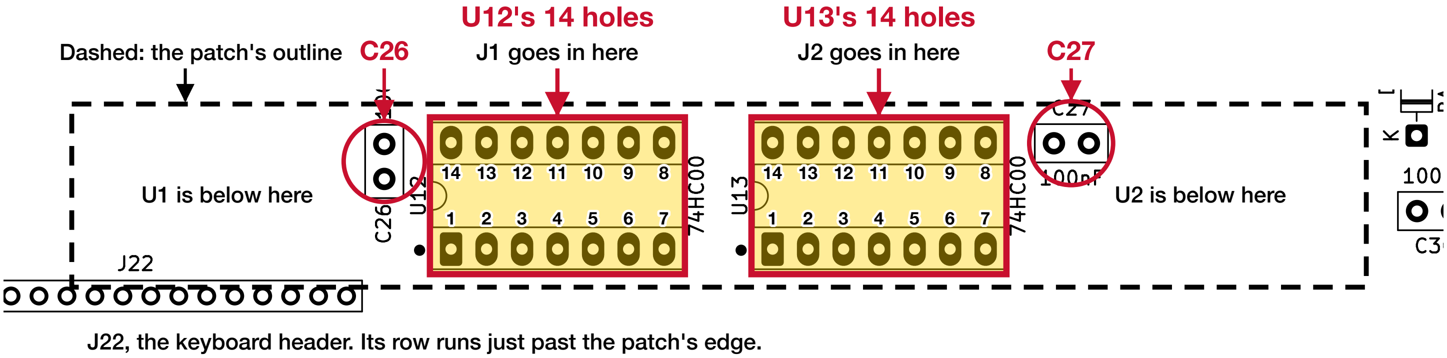

U12's 14 holes
J1 goes in here
U13's 14 holes
J2 goes in here
Dashed: the patch's outline
C26
C27
U1 is below here
U2 is below here
14 13 12 11 10 9 8
1 2 3 4 5 6 7
U12
74HC00
U13
74HC00
C26
C27
100nF
J22
100
K
C3
J22, the keyboard header. Its row runs just past the patch's edge.

Figure 9. Top of the ACE, keyboard toward you. The patch's four parts run in one row: U1, J1, J2, U2. J1 goes in U12's holes and J2 goes in U13's. Pin numbers are marked on both sockets.

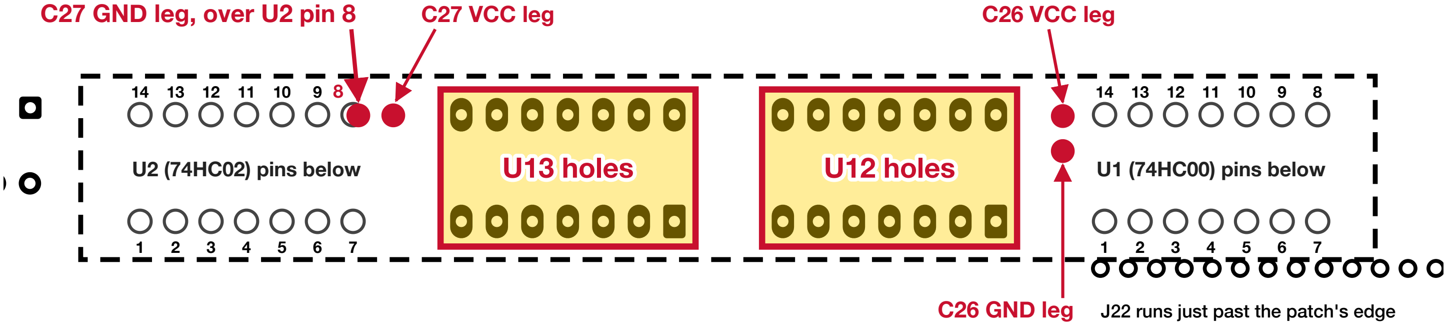

C27 GND leg, over U2 pin 8
C27 VCC leg
C26 VCC leg
14 13 12 11 10 9 8
U2 (74HC02) pins below
1 2 3 4 5 6 7
U13 holes
U12 holes
U1 (74HC00) pins below
C26 GND leg
J22 runs just past the patch's edge

Figure 10. Underside of the ACE, keyboard toward you. Left and right swap, so U13 is left of U12. Red dots are the legs of C26 and C27. Numbered white circles show where the patch's U1 and U2 pins land, 2.54 mm below the ACE.

Watch C27: its GND leg sits 0.38 mm off the center of U2 pin 8, with only the 2.54 mm header gap between them. U2 pin 8 carries the signal from U1 pin 8. If the leg touches the joint, that signal is shorted to ground. Step 2 deals with this.

1 Try the header pins in the holes

Do this before soldering anything. Both boards have 0.8 mm holes at these pins. A standard 0.64 mm square pin measures about 0.9 mm across its corners, so it may be tight. Push one strip through J1's row on the patch and through U12's row on the ACE, by hand, then take it out.

☐ The pins go in with finger pressure: use the strips.

☐ They need force: stop, and use round machined-pin header instead. Forcing a pin can damage the plated hole. Check that the chips still clear the plate (Figure 13).

2 Clear the underside of the ACE

- ☐ **C27 and C26:** clip all four legs to no more than 1 mm below the ACE. C27's GND leg matters most.
- ☐ The whole strip inside the dashed outline in Figure 10 is free of wires and parts.
- ☐ **J22:** its tails stand just past the patch's long edge, about 0.3 mm clear. Keep them straight.

3 Solder U1 and U2 to the patch

Hold the patch with its text side toward you and the text upright. This is the back. The chips go on this side.

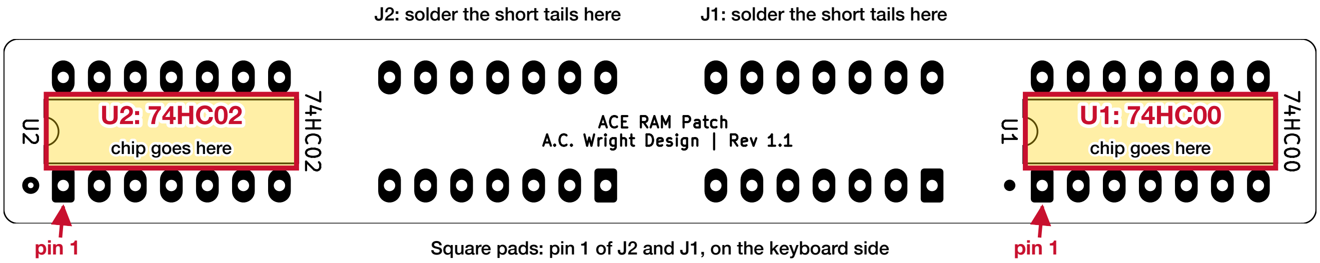

J2: solder the short tails here
J1: solder the short tails here
U2
U2: 74HC02
chip goes here
74HC02
ACE RAM Patch
A.C. Wright Design | Rev 1.1
U1
U1: 74HC00
chip goes here
74HC00
pin 1
Square pads: pin 1 of J2 and J1, on the keyboard side
pin 1

Figure 11a. The patch's back: text upright. U2 (74HC02) is on the left and U1 (74HC00) on the right. Both have pin 1 at lower left, so both notches point left.

- ☐ U2, the **74HC02**, on the left. U1, the **74HC00**, on the right. Check the markings: the two chips look alike.
- ☐ Notches to the left. Chips seated flat on the board. **No sockets:** a socket adds about 3 mm and the chip would hit the plate.
- ☐ Solder all 28 legs on the front. Clip them flush now: this side faces the ACE and you can't reach it later.

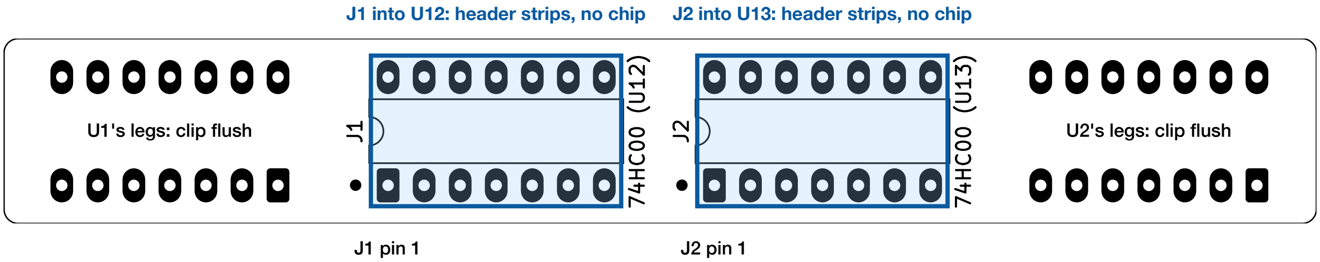

J1 into U12: header strips, no chip
J2 into U13: header strips, no chip
U1's legs: clip flush
J1
74HC00 (U12)
J2
74HC00 (U13)
U2's legs: clip flush
J1 pin 1
J2 pin 1

Figure 11b. The patch's front. The silkscreen on J1 and J2 looks like two chip outlines. Those are headers. No chip goes on this side.

4 Stand the header strips in the ACE

- ☐ Turn the ACE face down, keyboard edge toward you. Hold it up on blocks at the board's edges, so nothing rests on any keycaps or chips and the pins have room below.
- ☐ Push four 7-pin strips into U12's and U13's holes from the underside: **long ends down, through the ACE.** The plastic sits flat on the ACE, with the short tails pointing up at you.

5 Lower the patch onto the strips, and check which way round it is

☐ Patch with chips up, so the front faces the ACE. The short tails come up through J1 and J2. The patch rests on the plastic.

The patch also fits turned 180°, and that is wrong. Every pin finds a hole, but VCC lands on GND (J1 pin 14 goes in U13 pin 7). Check all of these before any solder:

☐ Keyboard edge toward you, and the text on the patch reads **upright**.
☐ U2 (74HC02) on your left and U1 (74HC00) on your right.
☐ The patch's square J1 and J2 pads are on the keyboard side, under the ACE's square U12 and U13 pads.
☐ C27's and C26's legs don't touch the patch. Look through the gap with a light.

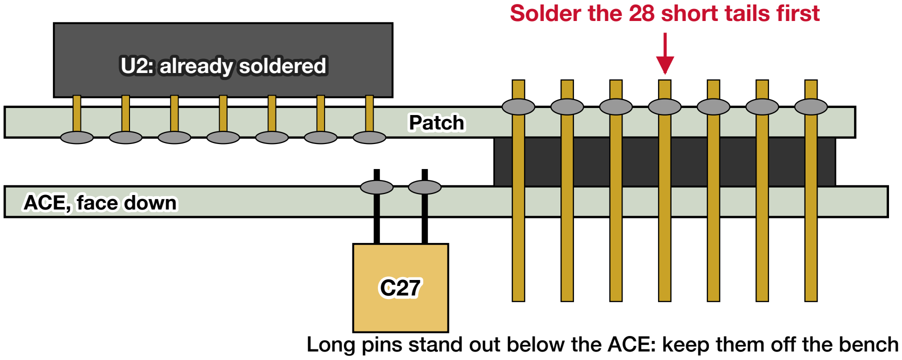

Solder the 28 short tails first
U2: already soldered
Patch
ACE, face down
C27
Long pins stand out below the ACE: keep them off the bench

Figure 12. Steps 4 to 6: the ACE face down, seen from the keyboard edge (one row of pins, to scale). U2 is on the left in this view.

6 Solder the short tails on the patch's back

☐ On each of the four strips, tack the two end pins only.
☐ Check that the patch lies flat on all the plastic, and the plastic flat on the ACE. Reheat a tack to correct it.
☐ Solder the rest: 28 joints in all, including the pins that show no track. They hold the patch on.

7 Solder the long pins on top of the ACE

☐ Turn the ACE face up. Hold it on blocks or its standoffs, so the patch hangs free. It hangs about 8.5 mm.
☐ Push the patch up, so the plastic is against the ACE's underside. Tack the end pins of each strip and check again.
☐ Solder the rest: 28 joints in U12's and U13's pads. Clip the pins to about 1 mm above the joint.

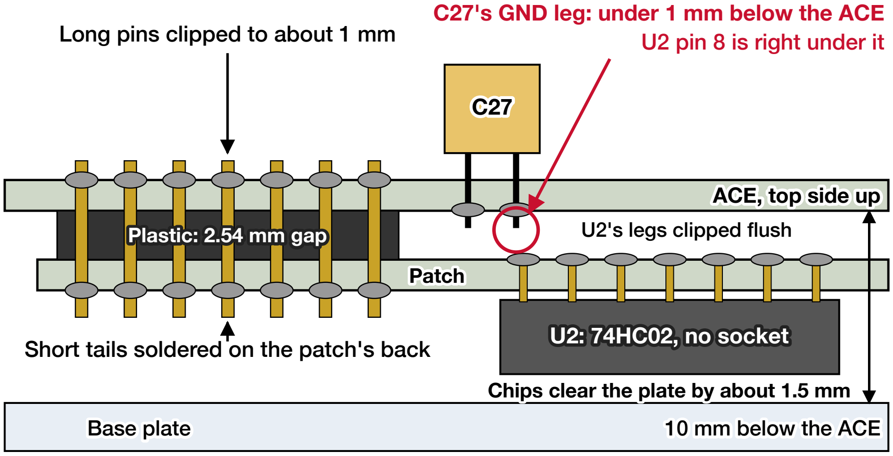

Long pins clipped to about 1 mm
C27's GND leg: under 1 mm below the ACE
U2 pin 8 is right under it
C27
ACE, top side up
Plastic: 2.54 mm gap
U2's legs clipped flush
Patch
U2: 74HC02, no socket
Short tails soldered on the patch's back
Chips clear the plate by about 1.5 mm
Base plate
10 mm below the ACE

Figure 13. The finished stack, ACE on top, seen from the keyboard edge (one row of pins, to scale).

Check the board with the meter, with no power on

Use the continuity beep. Figure 10 numbers the U1 and U2 pins, and Figure 9 numbers the U12 and U13 holes. Touch U1 and U2 at the chips' legs, from under the ACE. Inspect every joint first: 28 on the patch's back, 28 on top of the ACE, and U1's and U2's on the patch's front, seen through the gap with a light.

From	To	What this checks	Expect
J19 pin 1	J19 pin 2	VCC and GND not shorted	**No steady beep.** A short chirp is fine
U1 pin 14	C26 VCC leg	Patch has power; not turned 180°	Beep
U1 pin 7	C26 GND leg	U1 has ground	Beep
U2 pin 14	C27 VCC leg	U2 has power	Beep
U2 pin 7	C27 GND leg	U2 has ground	Beep
U2 pin 8	C27 GND leg	C27's leg not touching U2 pin 8	**No beep**
U2 pin 10	U13 pin 8, ACE top	LOADL reaches the ACE	Beep
U2 pin 13	U13 pin 11, ACE top	LOADH reaches the ACE	Beep
U12 pin 1, ACE top	U12 pin 2, ACE top	A9 and A8 not bridged	**No beep**

Any row wrong: do not power the ACE. A beep from U1 pin 14 to C26's GND leg means the patch is turned 180°.

First power-on

- ☐ SW69, IO ENABLE: all eight switches on. Each switch is one I/O window, IO1 to IO8, and the first two are the banked RAM.
- ☐ A center-positive 5 V supply. Switch on.

With a pico9918 running 6502-PICOVDP and BIOS 2.x, the screen shows the AC6502 logo, then `AC6502 BIOS v2.0`, `BASIC v2.0 30718 BYTES FREE` and the cards the BIOS found: `RAM RTC CF SER VIA SID VDP`. The 6502-ACE README's "Video card" section covers setting up the card and the ROM.

Which cards did the BIOS find?

Type this at BASIC:

```
PRINT PEEK(781)
```

With everything fitted it prints **255**. Each card the BIOS doesn't find takes its own value off that total:

RAM, $8000	RAM, $8400	RTC	CF	Serial	VIA	SID	Video
1	2	4	8	16	32	64	128

Without the patch, the BIOS doesn't find the $8400 window, so a full Rev 1.0 prints 253. At 1 MHz the BIOS can miss an ARMSID on about one boot in a dozen: if the total is 64 short, press reset and read it again.

Does the banked RAM work?

Type this short program. It writes each bank of both windows and reads it back. Then it writes to the addresses that wrongly loaded the bank register on an unpatched board, and checks that the bank didn't move. It takes about a minute at most.

```
NEW
10 E=0
20 FOR W=0 TO 1
30 D=32768+1024*W:R=D+1023
40 FOR B=0 TO 255:POKE R,B:POKE D,B:NEXT
50 FOR B=0 TO 255:POKE R,B:IF PEEK(D)<>B THEN E=E+1
60 NEXT
70 POKE R,7:POKE D,165:POKE R,8:POKE D,90:POKE R,7
80 FOR A=255 TO 767 STEP 256:POKE D+A,0:NEXT
90 POKE D+768,0:POKE D+1022,0
100 IF PEEK(D)<>165 THEN E=E+1000
110 NEXT W
120 IF PEEK(32768)<>165 THEN E=E+1000
130 PRINT "ERRORS:";E
RUN
```

It should print **ERRORS: 0**. A count under 1000 means banks that don't hold their data. A count of 1000 or more means a write moved the bank: look again at the patch's joints and at the meter table.

Keyboard, serial and reset

- ☐ The keyboard types at BASIC. A dead keyboard with a pico9918 fitted points to change 6, the /INT diode.
- ☐ The reset button, SW70, restarts the machine to the same boot screen.
- ☐ Serial works at 19200 baud over a null-modem DB-9 cable. The ACE's port is wired as DTE. J2 (CTS EN) and J4 (DCD EN) both stay on ground unless you need the handshake.